A6 Data quality

Expectations, health checks and reconciliation between layers.

A6-L1 Data contracts with severity levels and structured results

Results are data, severity decides the action, and one run reports every problem.

```python
CheckName = str
Status = Literal["PASS", "WARN", "FAIL", "SKIP"]
Severity = Literal["CRITICAL", "HIGH", "MEDIUM", "LOW"]

@dataclass(frozen=True)
class CheckResult:
    check_name: CheckName
    status: Status
    severity: Severity
    expected: str
    actual: str
```

What does a contract check produce?	A structured record: check_name, status, severity, expected, actual. Never a sentence.
CRITICAL versus WARN	CRITICAL failures stop the pipeline by raising. WARN results are logged and the run continues.
Why evaluate every check before raising?	One run reports every problem, so the supplier fixes everything in one round trip.
Lakeflow Declarative Pipelines expectations	@dp.expect keeps and records, @dp.expect_or_drop drops, @dp.expect_or_fail stops the update.

A6-L2 Statistical checks that survive empty, single-row and zero data

The strong version returns a result for every wave and every subgroup, and says why when a statistic is undefined.

F.stddev on one row, and mean on an empty frame	Both NULL. stddev is the sample standard deviation.
Division by zero on Spark 4.0 with ANSI on	Raises DIVIDE_BY_ZERO, even for DOUBLE. Use F.try_divide or F.when(b > 0, a / b).
What should a check return when a statistic is undefined?	An explicit SKIP result with the reason. Never nothing, and never a crash.
Why is `if cv:` wrong?	A real CV of 0.0 is falsy. Test missing values with `is None`.

A6-L3 Reconcile counts and totals between layers

Silver and gold totals are computed the same way and compared with tolerance. A mismatch becomes a CRITICAL result with numbers, not a print.

Bronze figure that must match silver's row count for a wave	Distinct (respondent_id, wave_id) keys, not raw bronze rows, because bronze keeps every delivery.
Silver-to-gold reconciliation	In-scope silver weight total = sum of gold weighted_base; in-scope silver rows = sum of unweighted_n.
How to compare floating-point totals	math.isclose(a, b, rel_tol=1e-9, abs_tol=1e-6). Never ==.

Severity of a failed reconciliation	CRITICAL. Every figure for the wave is suspect until it passes.